

- 8 Elk are one of the largest species of deer. The photograph shows three elk grazing in Yellowstone National Park.

They prefer to feed on young tree shoots, such as aspen and cottonwoods, rather than grass.



(Source: ©Westend61 GmbH/Alamy Stock Photo)

- (a) Wolves were introduced into Yellowstone National Park in 1995.

Wolves are predators of elk.

Before the introduction of wolves, elk grazed undisturbed, eating all the young tree shoots in the forest and riverside habitats.

The table shows how the percentage of young shoots eaten by the elk changed after the introduction of wolves.

Year	Percentage of young tree shoots eaten by elk (%)	
	Forest habitat	Riverside habitat
1996	100	100
1998	100	100
2000	100	92
2002	90	81
2004	86	45
2006	63	11
2010	19	2



Explain why the percentage of young tree shoots eaten by the elk changed after the introduction of wolves.

(3)

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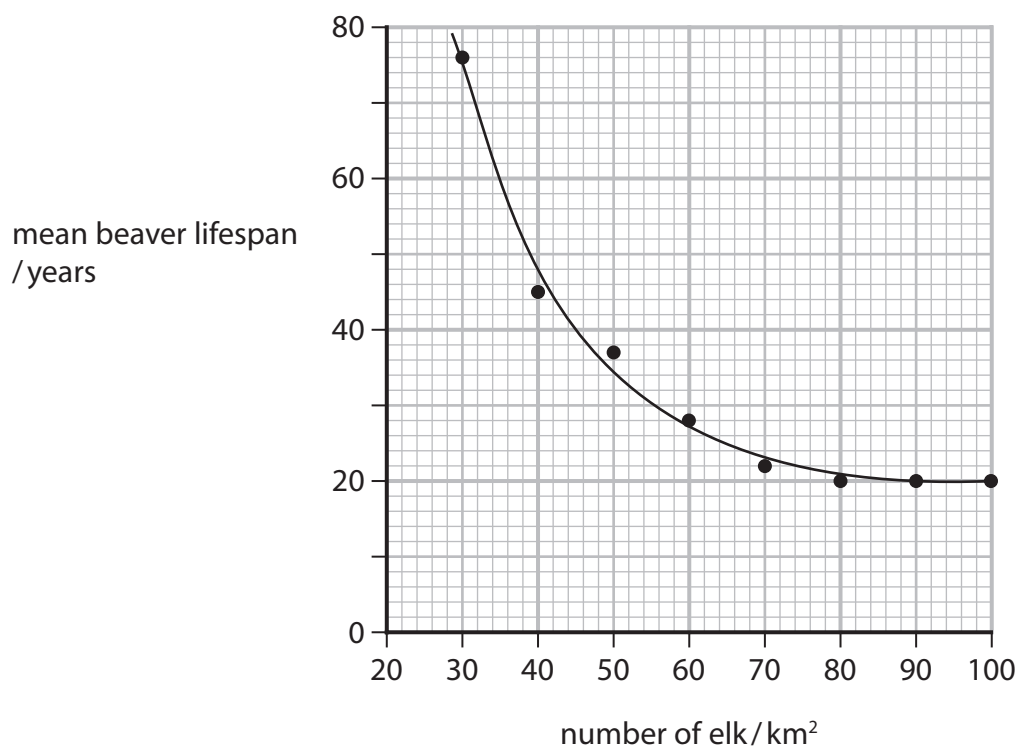
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(b) The photograph shows a beaver.



(Source: © Ian MacNicol/Getty Images)

The graph shows the mean beaver lifespan in areas with different numbers of elk per km^2 .



Describe the relationship shown in the graph.

(3)

(c) Scientists investigated the effect of the introduction of wolves on the biodiversity of Yellowstone National Park.

(i) Suggest what information the scientists would need to collect in order to calculate species richness in a habitat in Yellowstone National Park.

(2)

(ii) Write a formula that could be used to determine the biodiversity of a habitat in Yellowstone National Park.

(1)

*(d) Beavers and elk eat the shoots of young trees.

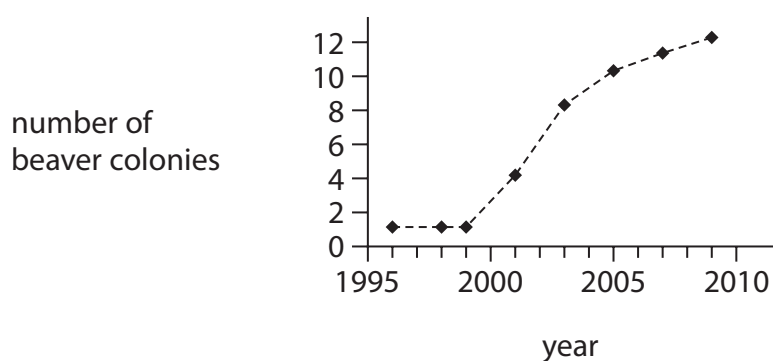
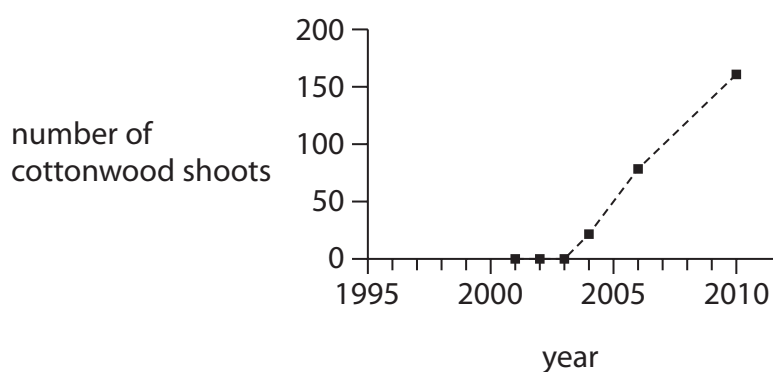
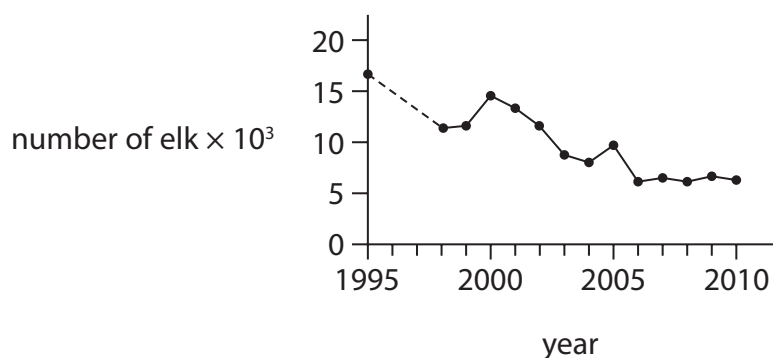
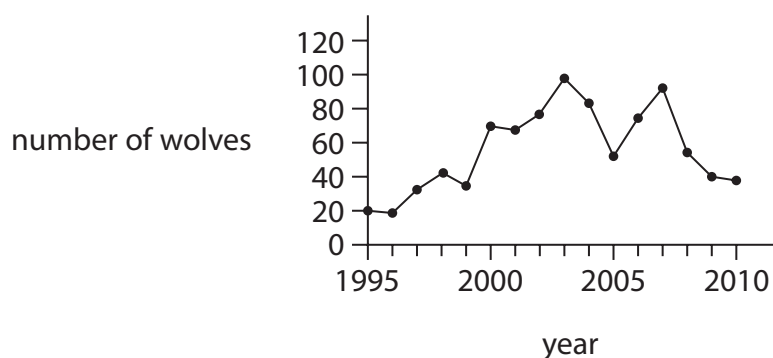
Beavers use older trees to build dams in the rivers, which create new habitats, such as ponds.

Beavers live in colonies of up to 8 individuals.

After the introduction of the wolves, scientists recorded the number of wolves, elk and beaver colonies for 15 years.

They also recorded the number of young cottonwood shoots that were taller than 5 cm.

The graphs show the data collected by these scientists.



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Discuss the effect of the introduction of wolves on the biodiversity of Yellowstone National Park.

Use all the information in the question to support your answer.

(6)

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(Total for Question 8 = 15 marks)

TOTAL FOR PAPER = 80 MARKS



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